Mathematics-1 for Data Science

Week 3 Notes

Course: Mathematics–1

Notes created by: Vishal (Instructor)

IIT Madras BS Degree Programme

Contents

Lecture	Topic	Page
Lecture 1	Quadratic Functions	2
Lecture 2	Applications of Quadratic Functions	8
Lecture 3	Slope of a Quadratic Function	12
Lecture 4	Quadratic Equations and Their Roots	17
Lecture 5	Slope by the Symmetric Difference Formula	22
Lecture 6	Quadratic Functions: A Summary	26
Lecture 7	Solving by Factoring	30
Lecture 8	Completing the Square	35
Lecture 9	The Quadratic Formula and the Discriminant	38
Lecture 10	Representations of a Quadratic Function	44

Lecture 1 — Quadratic Functions

The parabola, its axis of symmetry and vertex, and maximum and minimum values

What this lecture covers

We begin a new topic: **quadratic functions**. We started the course with simple geometric objects — points and lines — plotted them on the coordinate plane, and derived the algebraic equation of a line, arriving at $y = mx + c$, the **linear function**. Now we add one more layer of complexity. Here we take the *opposite* approach to the one used for straight lines: we first state the algebraic form and then derive its geometric properties.

1 Defining a Quadratic Function

Quadratic function

A **quadratic function** is described by an equation of the form

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

Why $a \neq 0$ is crucial

If $a = 0$ the equation reduces to $bx + c$ — a **linear** function, which as a geometric object is a straight line. So the condition $a \neq 0$ is what makes it quadratic.

Where the name comes from

The name comes from the Latin *quadratum*, meaning **square**, so *quadratic* means *related to the square*. A quadratic function is one related to the square of the variable — as the definition shows, it has a term containing ax^2.

The three terms

The equation is composed of three parts:

ax^2	bx	c
quadratic term	**linear term**	**constant term**

- When $a = 0$, only $bx + c$ survives — and that we already know how to handle: it is the equation of a non-vertical line.
- Putting $x = 0$, the only term that survives is c — the constant term.

So what happens when the ax^2 term remains, that is, when $a \neq 0$?

Parabola

The graph of any quadratic function is called a **parabola**.

2 Plotting a Parabola

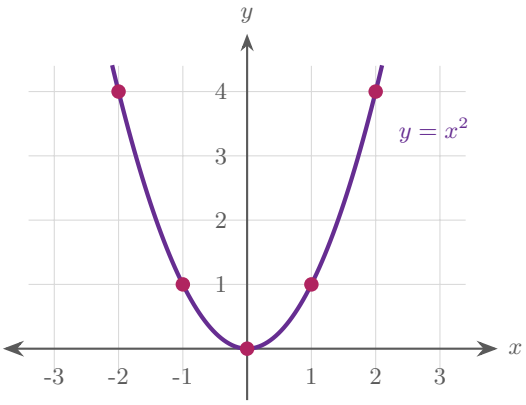
To graph any function we take values of x, put them into the formula $f(x)$, evaluate, and plot the resulting ordered pairs on the coordinate plane.

The simplest case — $y = x^2$

Here $a = 1$, $b = 0$, $c = 0$:

$$x = 0 \;\to\; y = 0, \qquad x = \pm 1 \;\to\; y = 1, \qquad x = \pm 2 \;\to\; y = 4$$

Joining these points smoothly gives the curve $y = x^2$.



The recipe for drawing a quadratic function

1. Generate a **table** of ordered pairs satisfying the function.
2. **Plot** those points on the coordinate plane.
3. Connect them with a **smooth curve**.

An example

Graphing $f(x) = x^2 + 2x + 1$

Computing a few values — for instance at $x = -2$:

$$(-2)^2 + 2(-2) + 1 \;=\; 4 - 4 + 1 \;=\; 1$$

and at $x = -1$:

$$(-1)^2 + 2(-1) + 1 \;=\; 1 - 2 + 1 \;=\; 0$$

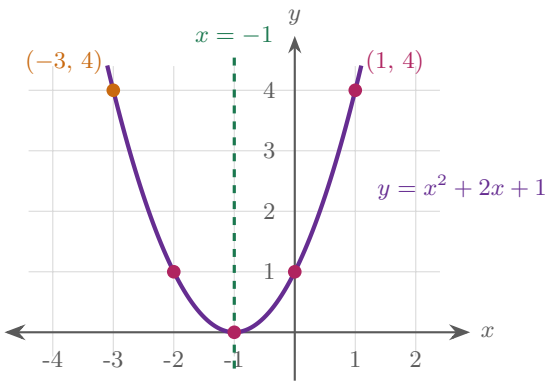
x	-2	-1	0	1
$f(x)$	1	0	1	4

Plotting these points and joining them smoothly gives the curve. But how do we know the shape in the region to the *left*, where we have no points?

The hidden symmetry

The parabola is symmetric. Take the point $(1, 4)$. The point $x = -1$, where the curve takes the value 0, is 2 units away from $x = 1$. So at $x = -3$ — which is 2 units away from $x = -1$ on the other side — the parabola will again take the value 4.

Indeed, drawing the straight line $x = -1$ through the bottom of the curve: for every point on one side there is a matching point on the other at the same distance from that line.



3 Axis of Symmetry, Vertex and y-Intercept

Axis of symmetry

Every parabola $y = ax^2 + bx + c$ has an **axis of symmetry**: if we take a graph paper containing the parabola and fold it along that line, the two portions of the curve match each other exactly.

This is the beauty of a parabola — knowing how it behaves on one side tells us how it behaves on the other, since that side is a pure reflection.

Vertex and y-intercept

The point at which the axis of symmetry meets the parabola is called the **vertex**.
Putting $x = 0$ in $ax^2 + bx + c$ leaves only c, so the **y-intercept** is

$$c.$$

These three things — the y-intercept, the axis of symmetry and the vertex — play a crucial role in graphing a parabola.

$$\text{axis of symmetry: } \quad x = -\frac{b}{2a} \qquad\qquad \text{vertex } x\text{-coordinate: } \quad -\frac{b}{2a}$$

On this formula

The derivation needs algebraic skills we do not have yet, so it comes later; for now, memorise $x = -\dfrac{b}{2a}$. Note that c plays *no* role in it — only a and b do. And since the axis of symmetry is $x = -\dfrac{b}{2a}$, the x coordinate of the vertex is the same.

y-intercept
c

axis of symmetry
$x = -\dfrac{b}{2a}$

vertex
$\left(-\dfrac{b}{2a},\ f\left(-\dfrac{b}{2a}\right)\right)$

4 A Fully Worked Graph

Graphing $f(x) = x^2 + 8x + 9$

y-intercept. Putting $x = 0$ gives 9.
Axis of symmetry. Here $a = 1$ and $b = 8$, so

$$x = -\frac{b}{2a} = -\frac{8}{2} = -4.$$

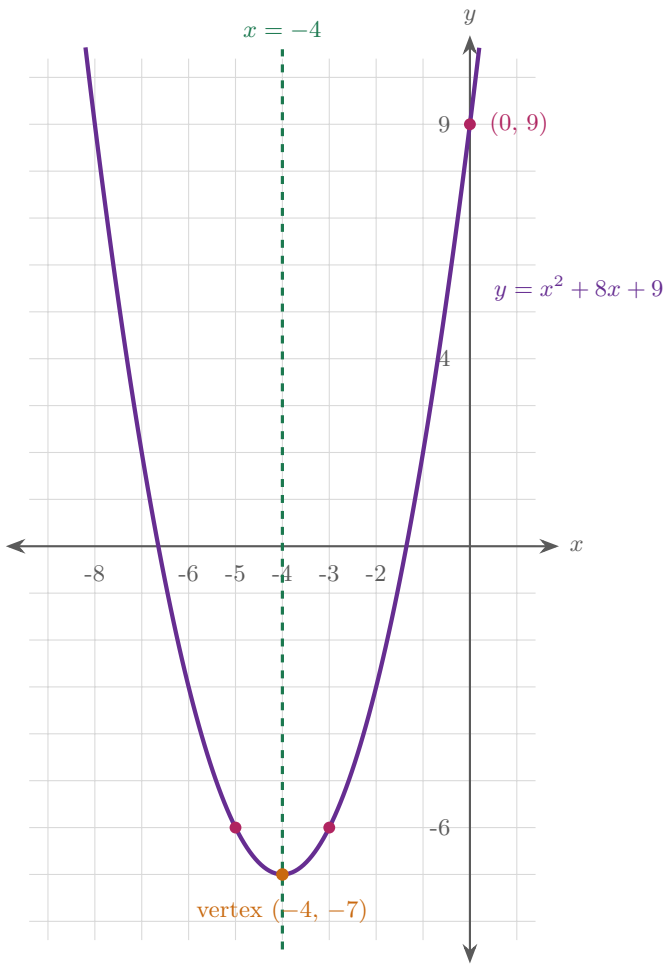
Vertex. Substituting $x = -4$ into the function:

$$(-4)^2 + 8(-4) + 9 = 16 - 32 + 9 = -7$$

so the vertex is $(-4,\ -7)$.

Knowing the vertex, we choose points *around* it rather than all on one side:

x	-5	-4	-3	0
$f(x)$	-6	-7	-6	9



Knowing the axis of symmetry tells us exactly where the minimum occurs, which is why the graph can now be drawn with ease.

5 Parabolas That Open Downwards

Is the upward shape the only one possible? Consider

$$f(x) \;=\; -x^2 + 1.$$

Is it quadratic, and what are its three features?

Yes — here $a = -1$, $b = 0$, $c = 1$.

$$y\textbf{-intercept} \;=\; 1$$

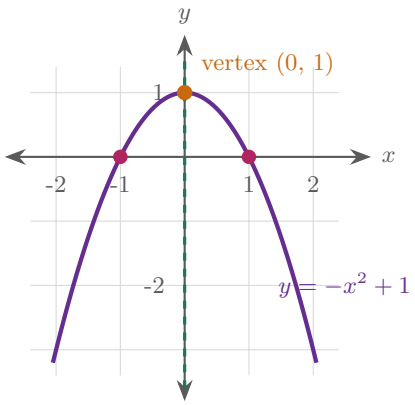
$$\textbf{axis of symmetry:} \quad x = -\frac{b}{2a} = 0 \qquad (\text{since } b = 0, \text{ whatever } a \text{ may be})$$

so the Y axis itself is the axis of symmetry, and the **vertex** is $(0,\ 1)$.

When the three features coincide

Here we do not gain much extra information: $(0, 1)$ is the y-intercept, the axis of symmetry is $x = 0$, so $(0, 1)$ is the vertex as well. Still, this suffices — we know to take points *around* $x = 0$.

x	-1	0	1
$f(x)$	0	1	0



Here the curve opens **down** rather than up. Why? In the earlier examples $a = 1$, whereas here $a = -1$ — and that is precisely what flips the curve.

6 Maximum and Minimum Values

The key fact

The y coordinate of the **vertex** of a quadratic function is the **minimum** or **maximum** value attained by the function.

This agrees with everything we have seen: $y = x^2$ bottoms out at 0; $y = x^2 + 2x + 1$ again gives 0 at its lowest point; and $y = -x^2 + 1$, with a negative, goes downwards and so its vertex value is a **maximum**, with all other values below it.

$$a > 0 \implies \text{opens } \textbf{upwards}, \text{ has a } \textbf{minimum} \text{ value}$$

$$a < 0 \implies \text{opens } \textbf{downwards}, \text{ has a } \textbf{maximum} \text{ value}$$

Either, not both

A quadratic function has *either* a maximum *or* a minimum — never both. That is part of the beauty of the quadratic function.

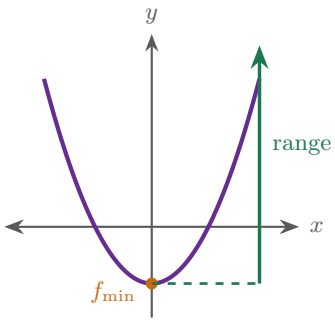
The range

Relating this to the domain–codomain–range discussion from Week 1: the **domain** of a quadratic function is the entire real line, while the **range** is restricted to a subset of it.

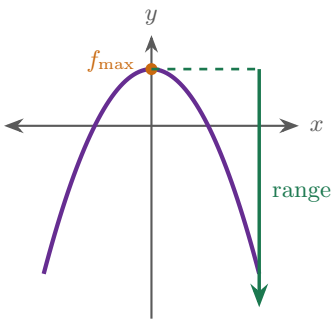
Range of a quadratic function

$$a > 0: \quad \mathbb{R} \cap \{ f(x) \mid f(x) \geq f_{\min} \} = [f_{\min}, \infty)$$

$$a < 0: \quad \mathbb{R} \cap \{ f(x) \mid f(x) \leq f_{\max} \} = (-\infty, f_{\max}]$$



$a > 0$ — minimum



$a < 0$ — maximum

Lecture 2 — Applications of Quadratic Functions

Finding maxima and minima, a real-world optimisation problem, and the road to slope

What this lecture covers

We put the previous lecture to work. First a straightforward example of locating a minimum and stating the domain and range; then a realistic optimisation problem about a tour bus, solved entirely with quadratic functions; and finally we begin asking what the notion of **slope** — so familiar for a straight line — should mean for a quadratic function.

1 A Worked Example

The problem

Given

$$f(x) \;=\; x^2 - 6x + 9,$$

determine whether f has a minimum or a maximum value; if so, find that value; and state the domain and range of f.

Domain

The domain of f is the **entire real line** — there is nothing to worry about here.

Minimum or maximum?

Identify the coefficients:

$$a \;=\; 1, \qquad b \;=\; -6, \qquad c \;=\; 9.$$

Since $a > 0$, the function **opens up**, and a function that opens up has a **minimum** value. Once it has a minimum it cannot also have a maximum.

The value

The x coordinate of the vertex is

$$x \;=\; -\frac{b}{2a} \;=\; -\frac{-6}{2\times 1} \;=\; \frac{6}{2} \;=\; 3$$

Substituting back:

$$f(3) \;=\; 3^2 - 6(3) + 9 \;=\; 9 - 18 + 9 \;=\; 0$$

so the minimum value is **0**, attained at the vertex $(3,\ 0)$.

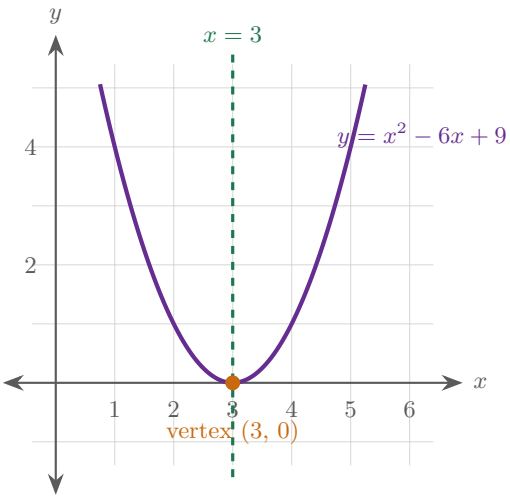
Mind the two minus signs

Note carefully that $-\dfrac{b}{2a}$ with $b = -6$ gives $-\dfrac{(-6)}{2} = +3$, not -3 — the two minus signs cancel.

Range

Since the function has a minimum, the range runs from that minimum upwards — the entire real line above it:

$$\text{range} \;=\; \mathbb{R} \cap \{\, f(x) \;\mid\; f(x) \geq 0 \,\} \;=\; [0,\ \infty)$$



If instead a had been negative, we would similarly have found a **maximum** value.

2 A Realistic Application: The Tour Bus

The problem

A tour bus in Chennai serves 500 customers per day and charges Rs. 40 per person. The owner estimates that the company would lose 10 passengers per day for each Rs. 4 hike in the fare. The company wants to maximise its income. **How much should the fare be?**

Setting up the variable

One unit of hike is Rs. 4. Let

$$x \;=\; \text{the number of Rs. 4 fare hikes.}$$

$$\text{fare per passenger} \;=\; 40 + 4x \qquad\qquad \text{number of passengers} \;=\; 500 - 10x$$

The fare rises by Rs. 4 for each unit of x, while 10 passengers are lost for each unit, so $500 - 10x$ passengers remain.

The income function

Income is the number of passengers times the fare charged:

$$\begin{aligned} I(x) \;&=\; (500 - 10x)(40 + 4x) \\ &=\; 20000 + 2000x - 400x - 40x^2 \;=\; -40x^2 + 1600x + 20000 \end{aligned}$$

Is a maximum possible?

Yes — and the answer lies in the coefficients:

$$a = -40, \qquad b = 1600, \qquad c = 20000.$$

Since $a = -40 < 0$, the curve opens **downwards**, so a maximum exists.

Where is it attained?

The maximum is attained at the vertex, whose x coordinate is the point where the axis of symmetry meets the curve:

$$x = -\frac{b}{2a} = -\frac{1600}{2 \times (-40)} = \frac{1600}{80} = 20$$

the minus signs cancelling. The maximum income is therefore

$$I(20) = -40(20)^2 + 1600(20) + 20000 = -16000 + 32000 + 20000 = \text{Rs. } 36{,}000$$

Compare with the present income

At present the company serves 500 customers each paying Rs. 40 — that is $x = 0$:

$$I(0) = 20000$$

so it earns only Rs. 20,000.

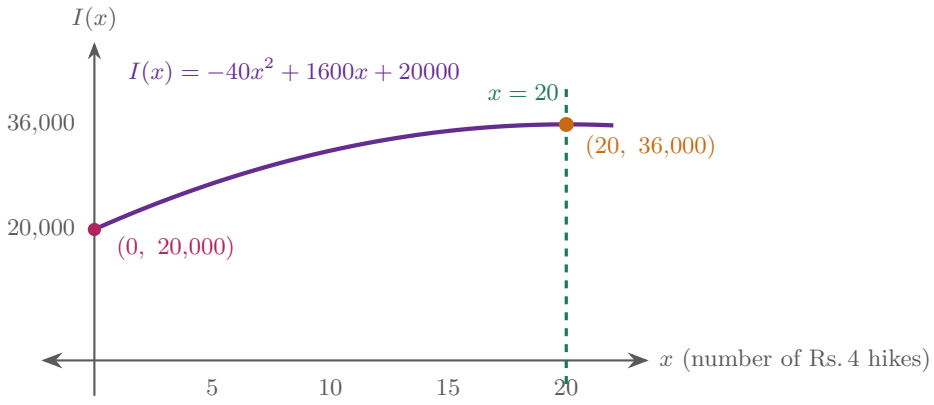
The recommendation

The solution says there should be 20 units of hike at Rs. 4 each — that is, a hike of Rs. 80. The new fare is

$$40 + 4x = 40 + 4(20) = 40 + 80 = \text{Rs. } 120$$

So each person should be charged Rs. 120 instead of Rs. 40. The company becomes more profitable while serving fewer customers.

	$x = 0$	$x = 20$
Fare per passenger	Rs. 40	Rs. 120
Number of passengers	500	300
Income	Rs. 20,000	Rs. 36,000



This is how quadratic functions are used to solve real-life situations.

3 Towards the Slope of a Quadratic Function

Let us go back to linear functions, where we studied the slopes of lines. What was the slope of a line? It was the change in y over the change in x. What does that concept have to do with a quadratic function?

The goal

Given a quadratic function

$$f(x) = ax^2 + bx + c, \qquad a \neq 0,$$

how do we determine the **slope** of f?

What we know for a linear function

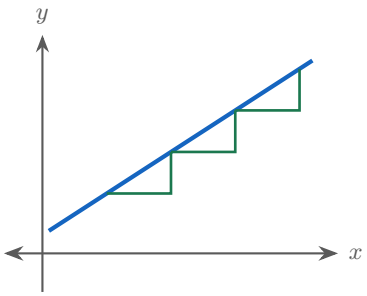
For a linear function

$$y = g(x) = mx + c$$

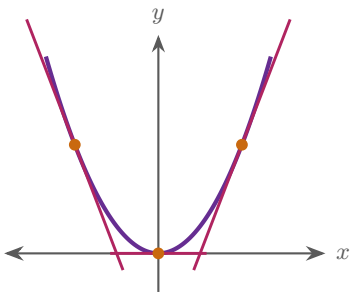
we know that m is the slope, and that it can be calculated as

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{\Delta y}{\Delta x} = \tan\theta$$

where θ is the inclination with the positive X axis. Crucially, for a line the slope remains **constant** everywhere.



line
slope is the same everywhere



parabola
steepness changes from point to point

The question ahead

We shall use a similar analogy to define the slope of a quadratic function. To discuss the concept, we start with our standard prototype example:

$$y = x^2.$$

Lecture 3 — Slope of a Quadratic Function

From chords to tangents, second differences, and the result $2ax + b$

What this lecture covers

We want to know what the **slope** of a parabola is — and whether it is a constant, a variable, or something else. Starting from the prototype $y = x^2$, we compute differences, discover that they are the slopes of *chords* rather than of the curve, and use the inclination idea to attach each chord's slope to a definite point. The pattern generalises to any quadratic and yields the slope $2ax + b$.

1 The Prototype $y = x^2$

First we plot the function. Here $b = 0$, so the Y axis is the axis of symmetry, the curve is symmetric about it, and the minimum value is 0.

Taking points symmetric about $x = 0$:

x	-2	-1	0	1	2
$y = x^2$	4	1	0	1	4

The symmetry is clearly visible in these values.

Computing the differences

The definition of slope is the change in y over the change in x. Looking at the first column, the change in x is **constant** — it is 1 every time. Since the denominator is always 1, it suffices to take the differences of the y values.

$$\text{slope} = \frac{\Delta y}{\Delta x} = \frac{y_i - y_{i-1}}{1} = y_i - y_{i-1}$$

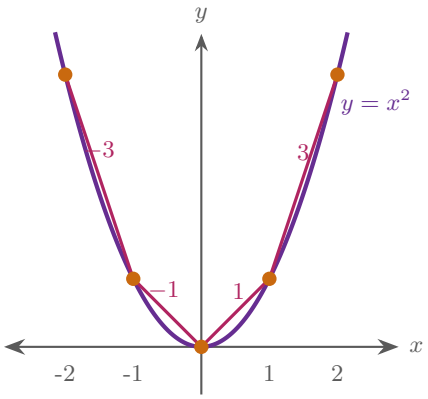
The first differences

$$1 - 4 = -3, \qquad 0 - 1 = -1, \qquad 1 - 0 = 1, \qquad 4 - 1 = 3$$

So we have obtained the changes in y corresponding to a one-unit change in x. But where does each of these slopes *live* — at what point? For a straight line the slope is constant, so the question does not arise; here it does.

2 These Are Slopes of Chords, Not of the Curve

When we take two points and compute $y_i - y_{i-1}$, we are effectively drawing a **straight line joining those two points** and calculating its slope. So this is the slope of that straight line — *not*, as it stands, the slope of the curve.



Bringing back the inclination

How then do we identify the slope of the *curve*? Recall the old definition: the change in y over the change in x is also $\tan\theta$, where θ is the angle of inclination. That definition never depended on a *particular* line — only on the inclination.

So if we pick a point on the curve, draw the line through it with that inclination, and measure the angle with the positive X axis, we get a slope. Such a line touches the curve at that point only — it is the **tangent**.

Matching a chord to a point

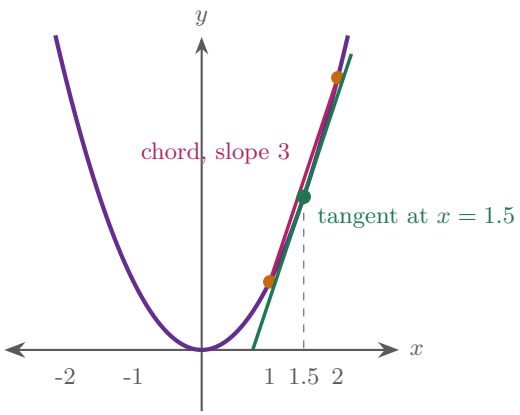
Take the chord between $x = 1$ and $x = 2$; its slope is 3. Sliding a line of the same inclination until it just touches the curve, it becomes the tangent at

$$x \;=\; 1.5$$

— and indeed

$$\frac{3}{1.5} \;=\; 2, \qquad \text{that is} \qquad 3 \;=\; 2 \times 1.5.$$

Note that 1.5 is exactly the **midpoint** of 1 and 2.



The second differences

Look now at the differences *of* the differences:

$$-1-(-3) \;=\; 2, \qquad 1-(-1) \;=\; 2, \qquad 3-1 \;=\; 2$$

All the **second differences** equal 2 — so there is a definite relation at work.

x	-2		-1		0		1		2
y	4		1		0		1		4
first difference		-3		-1		1		3	
attached at $x =$		-1.5		-0.5		0.5		1.5	
second difference			2		2		2		

Every chord matches its midpoint

$$\text{over } [1,2]: \text{ slope } 3 = 2(1.5) \qquad \text{over } [0,1]: \text{ slope } 1 = 2(0.5)$$
$$\text{over } [-1,0]: \text{ slope } -1 = 2(-0.5) \qquad \text{over } [-2,-1]: \text{ slope } -3 = 2(-1.5)$$

The slope of $y = x^2$

$$\text{slope} \;=\; 2x$$

Verifying the hypothesis

Take the point $x = 0$ and consider two points symmetric about it, namely 1 and -1. The line joining them is **horizontal**, so its slope is 0 — and $2(0) = 0$, exactly as predicted.

In general, for any point z we may take the two points

$$z + u \qquad \text{and} \qquad z - u,$$

join them by a straight line, and the slope of that line is the slope of the curve at z. This is a beautiful idea, and it generalises to any quadratic curve.

3 The General Quadratic

We now ask the general question: what is the slope of

$$f(x) \;=\; ax^2 + bx + c, \qquad a \neq 0\,?$$

Take the same five standard points $-2, -1, 0, 1, 2$ and substitute:

x	-2	-1	0	1	2
$f(x)$	$4a - 2b + c$	$a - b + c$	c	$a + b + c$	$4a + 2b + c$

First differences

$$(a - b + c) - (4a - 2b + c) \;=\; -3a + b \qquad c - (a - b + c) \;=\; -a + b$$
$$(a + b + c) - c \;=\; a + b \qquad (4a + 2b + c) - (a + b + c) \;=\; 3a + b$$

Second differences

Taking one more difference:

$$(-a + b) - (-3a + b) \;=\; 2a, \qquad (a + b) - (-a + b) \;=\; 2a, \qquad (3a + b) - (a + b) \;=\; 2a$$

so **all the second differences are** $2a$.

Attaching each to its midpoint

Each first difference belongs to the midpoint of its interval. For the interval $[-2,-1]$ the midpoint is -1.5, and

$$2a(-1.5)+b \;=\; -3a+b \qquad \checkmark$$

which is exactly the first difference computed above. Likewise $-a+b$ sits at -0.5, $a+b$ at 0.5, and $3a+b$ at 1.5.

Slope of a quadratic function

$$\boxed{\text{slope of } f(x)=ax^2+bx+c \;\text{ is }\; 2ax+b}$$

Reading the formula

The $2a$ comes from the second differences, and the x from the midpoint that each first difference is attached to — giving $2ax$ — with b carried along. For the prototype $y=x^2$, where $a=1$ and $b=0$, this reduces to $2x$, as we found.

4 Two Observations

Why the vertex sits at $x=-\frac{b}{2a}$

For $y=x^2$ we saw that around $x=0$ the slope was flat — it was 0. When does that happen in general? Set the slope to zero:

$$2ax+b \;=\; 0 \qquad\Longrightarrow\qquad x \;=\; -\frac{b}{2a}$$

Slope as a rate of change

The slope measures the **rate of change** of y with respect to x. If that rate of change becomes 0, the function has reached its minimum or maximum.

This is one reason why $x=-\dfrac{b}{2a}$ gives the minimum or maximum of a quadratic function: it is precisely where the slope reaches the value 0.

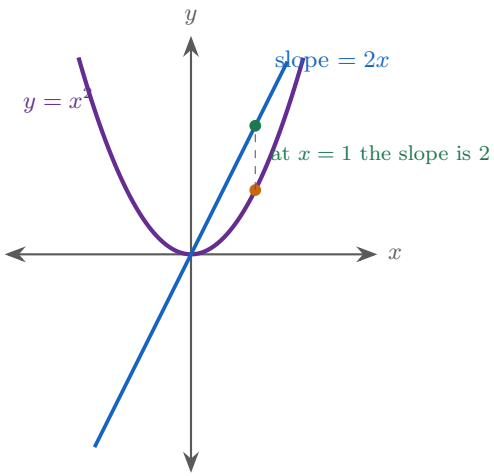
Still pending

Why the *axis of symmetry* is $x=-\dfrac{b}{2a}$ is a separate matter, and we shall come to it later.

Line versus parabola

	Line $mx+c$	Quadratic ax^2+bx+c
Slope	m	$2ax+b$
Behaviour	constant	variable
Depends on	m only	a and b, not c

The slope of a quadratic function is significantly different from the slope of a line: for a line it is constant, whereas for a quadratic it is no longer constant — it is a **variable**, namely $2ax+b$. It depends on a and b but not on the constant c, which is only to be expected.



the slope of the parabola at each x is given by the straight line $2x$

Lecture 4 — Quadratic Equations and Their Roots

Standard form, zeros as x-intercepts, and solving by the graphing technique

What this lecture covers

We learn what a **quadratic equation** is, what the solutions — called **roots** — mean, and how to find them using the graphing technique developed for quadratic functions.

1 Quadratic Equations

Quadratic equation

If a quadratic function is set equal to a value, the result is called a **quadratic equation**. For example

$$ax^2 + bx + c \;=\; 0, \qquad a \neq 0$$

is one quadratic equation, and

$$ax^2 + bx + c \;=\; 5, \qquad a \neq 0$$

is another.

The quantities a, b and c are called the **coefficients** of the quadratic equation.

Standard form

If the coefficients are integers and the right-hand side is 0, that is

$$ax^2 + bx + c \;=\; 0, \qquad a, b, c \in \mathbb{Z}, \qquad a \neq 0,$$

then the quadratic equation is said to be in **standard form**.

2 Roots and Zeros

Roots

The **solutions** of a quadratic equation are called its **roots**. That is, if

$$ax^2 + bx + c \;=\; 0,$$

the values of x making the left-hand side equal to 0 are the solutions of the equation, and those values are the roots.

How do we find them? One method is very easy: take the quadratic *function* associated with the equation, plot it, and find its **zeros**.

Zeros are x-intercepts

The zeros of a function are the **x-intercepts** of its graph, and these are exactly the solutions of the related equation $f(x) = 0$.

So: plot the function, and see where it meets the X axis. Wherever it does, we have a root.

3 Example 1 — Two Distinct Roots

Find the roots of $x^2 + 6x + 8 = 0$

The related quadratic function is $f(x) = x^2 + 6x + 8$, with

$$a = 1, \qquad b = 6, \qquad c = 8.$$

So the y-intercept is 8, and the axis of symmetry is

$$x \;=\; -\frac{b}{2a} \;=\; -\frac{6}{2} \;=\; -3.$$

Since $a > 0$, the graph **opens up** and attains a minimum. Substituting the axis value:

$$f(-3) \;=\; 9 - 18 + 8 \;=\; -1$$

which is **negative**. So the vertex lies below the X axis and the curve, opening up, must cross the axis at **two** points.

Guessing the two roots

Can we guess them without plotting? At $x = -3$ the value is negative, so try $x = -2$:

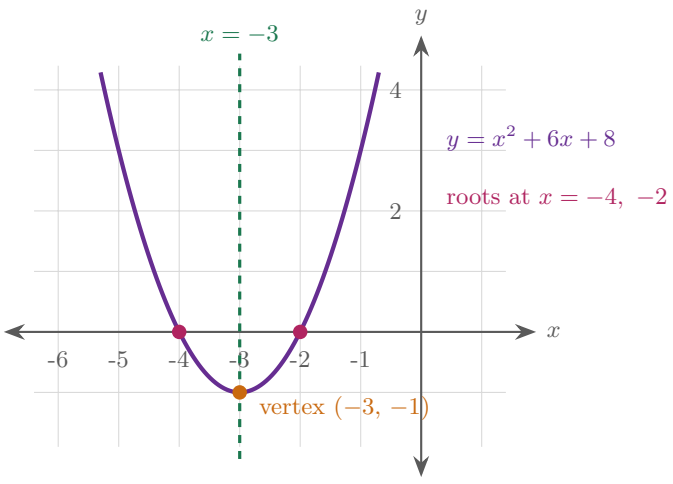
$$(-2)^2 + 6(-2) + 8 \;=\; 4 - 12 + 8 \;=\; 0$$

so -2 is one root. Since $x = -3$ is the axis of symmetry and -2 lies one unit away from it, there must be another root one unit away on the other side — namely $x = -4$:

$$(-4)^2 + 6(-4) + 8 \;=\; 16 - 24 + 8 \;=\; 0 \qquad \checkmark$$

roots: $x = -4$ and $x = -2$

So we found both roots purely from what we already understand — without drawing anything. The equation has **two real roots**.



4 Example 2 — One Repeated Root

Find the roots of $x^2 + 2x + 1 = 0$

The associated function is $f(x) = x^2 + 2x + 1$, with $a = 1$, $b = 2$, so

$$x = -\frac{b}{2a} = -\frac{2}{2} = -1$$

is the axis of symmetry.

Substituting $x = -1$:

$$f(-1) = (-1)^2 + 2(-1) + 1 = 1 - 2 + 1 = 0$$

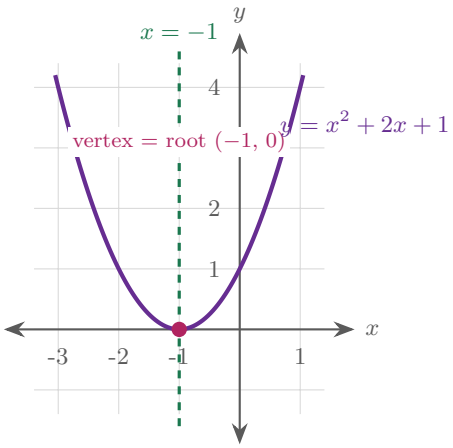
So -1 is itself a zero — but it is the **vertex**, where the minimum is attained.

Why there can be no other root

Since $a > 0$ the curve opens up and the vertex is its lowest point. If the curve already touches the X axis there, it rises on both sides and can never return to the axis. So by the axis of symmetry there is no other point where the value is 0:

$$\text{roots: } x = -1 \text{ and } x = -1$$

that is, **one real root, repeated twice**.



5 Example 3 — No Real Roots

Find the roots of $x^2 + 1 = 0$

Comparing with the standard form $ax^2 + bx + c$ gives

$$a = 1, \qquad b = 0, \qquad c = 1.$$

Since $b = 0$, the graph is symmetric about $x = 0$ — the Y axis. Since $a > 0$, the curve opens upwards and attains a minimum at the vertex.

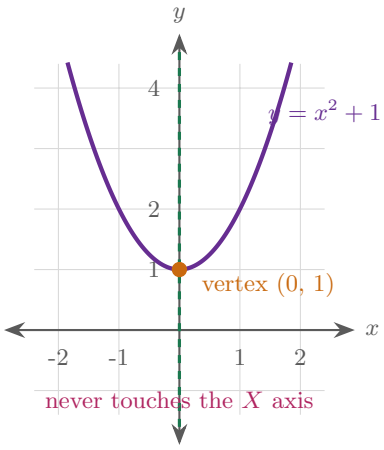
Substituting $x = 0$:

$$f(0) = 0^2 + 1 = 1$$

so the vertex is $(0,\ 1)$ and the minimum value of the function is 1.

Can the function ever be zero?

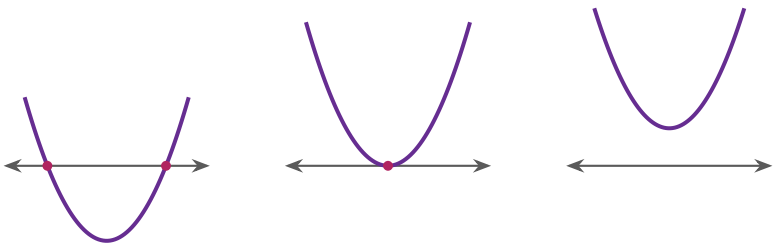
It cannot. The smallest value it ever takes is 1, which is above the X axis, so the curve **never intersects** the X axis. Therefore $x^2 + 1 = 0$ has **no real roots**.



6 All the Possible Cases

This summarises the possible outcomes for $ax^2 + bx + c = 0$, decided by the sign of a and by the y coordinate of the vertex.

Sign of a	Value at the vertex	Number of real roots
$a > 0$ (opens up)	negative	**two** real roots
$a > 0$	zero	**one** real root (repeated)
$a > 0$	positive	**no** real roots
$a < 0$ (opens down)	positive	**two** real roots
$a < 0$	zero	**one** real root (repeated)
$a < 0$	negative	**no** real roots



vertex below axis
two roots

vertex on axis
one root

vertex above axis
no real roots

The downward case

When $a < 0$ the curve opens downwards and attains its **maximum** at the vertex. If that maximum is positive there are two real roots; if it is zero there is a single repeated root; and if the whole curve lies below the X axis there are no real roots. This is worth working through

for yourself.

Lecture 5 — Slope by the Symmetric Difference Formula

Rate of change, the 2h formula, and deriving $2ax + b$ algebraically

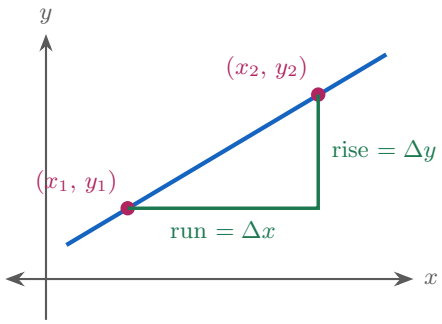
What this lecture covers

We revise the concept of the slope of a line and then **generalise** it to the slope of a quadratic function — this time algebraically rather than through tables of differences. The tool is a symmetric interval around a point, and the same formula handles lines and parabolas alike.

1 Slope of a Line, Revisited

The slope of a line is the **rise over run**: the rise is in the y direction and the run in the x direction. Given any two points (x_1, y_1) and (x_2, y_2) on the line:

$$\text{slope} = \frac{\text{rise}}{\text{run}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{\Delta y}{\Delta x}$$



Slope as sensitivity

The slope measures the **sensitivity** of the expression to a change in the x direction: how a small change in x affects the change in y. In other words, it measures the **rate of change**. Writing it as $\dfrac{\Delta y}{\Delta x}$ says that for a change Δx there is a corresponding change Δy.

For a **straight line** there is no change in the rate of change — the rate of change is **constant**. That very fact is what we used when deriving the equation of a (non-vertical) line,

$$y = mx + c \qquad \text{or, as a function,} \qquad f(x) = mx + c,$$

which is called a **linear function**.

Why generalise?

We want to handle the notion of rate of change for *all* functions, not just lines. For a straight line the answer is a constant and there is nothing to it — but what is the rate of change of a general function?

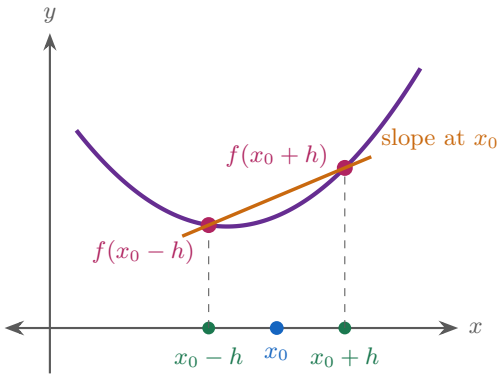
2 The Symmetric Interval

Suppose the slope were a *variable*. How would we then determine the slope at a particular point $(x_0,\ y_0)$?

Project the point onto the X axis at x_0, take a tiny interval of size h on each side, giving the two points

$$x_0 - h \qquad \text{and} \qquad x_0 + h,$$

and look at the corresponding points on the curve, $f(x_0 - h)$ and $f(x_0 + h)$. Then apply the rise-over-run formula to those two points.



Slope at a point

$$\text{slope at } x = x_0 \;=\; \frac{f(x_0 + h) - f(x_0 - h)}{(x_0 + h) - (x_0 - h)} \;=\; \boxed{\frac{f(x_0 + h) - f(x_0 - h)}{2h}}$$

Since the two points are taken symmetrically about x_0, the slope obtained is attached to the point x_0 itself.

3 Checking the Formula on a Line

Let us verify it for $f(x) = mx + c$. Substituting:

$$\text{slope} \;=\; \frac{[m(x_0 + h) + c] - [m(x_0 - h) + c]}{2h}$$

Expanding carefully:

$$= \frac{mx_0 + mh + c - mx_0 + mh - c}{2h}$$

The terms mx_0 cancel, c cancels with $-c$, and the two mh terms add:

$$= \frac{2mh}{2h} = m \qquad (h \neq 0)$$

As expected

The slope of a line at x_0 is m — which we already knew, since the rate of change of a line is always constant. So the formula is sound; it is simply more sophisticated and more robust, and we can now use it on a quadratic.

4 The Prototype $y = x^2$

Apply the same strategy to $f(x) = x^2$ at a point x_0:

$$\text{slope at } x_0 \;=\; \frac{(x_0 + h)^2 - (x_0 - h)^2}{2h}$$

Expanding both squares, keeping the brackets intact:

$$= \frac{\big[x_0^2 + h^2 + 2hx_0\big] - \big[x_0^2 + h^2 - 2hx_0\big]}{2h}$$

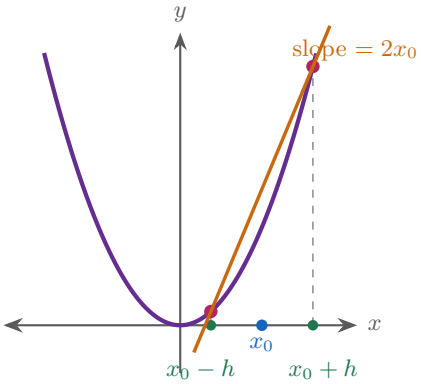
Now cancel: x_0^2 against x_0^2, and h^2 against h^2. The first bracket leaves $+2hx_0$, and the minus sign in front of the second bracket turns $-2hx_0$ into another $+2hx_0$:

$$= \frac{4hx_0}{2h} = 2x_0 \qquad (h \neq 0)$$

Slope of $y = x^2$

$$\text{slope at any point } x_0 = 2x_0 \qquad \text{that is} \qquad g(x) = 2x$$

which is itself another function.



5 The General Quadratic

Now let

$$f(x) = ax^2 + bx + c, \qquad a \neq 0.$$

The case $a = 0$

If $a = 0$ we already have the answer: the equation is that of a straight line, the slope is constant, and it equals b.

Using the same definition at any point x_0:

$$\text{slope at } x_0 = \frac{\big[a(x_0+h)^2 + b(x_0+h) + c\big] - \big[a(x_0-h)^2 + b(x_0-h) + c\big]}{2h}$$

Clubbing the terms

Group the a terms together, then the b terms, and note that $c - c$ cancels:

$$= \frac{a\big[(x_0+h)^2 - (x_0-h)^2\big] + b\big[(x_0+h) - (x_0-h)\big]}{2h}$$

For the first bracket we may reuse the calculation just done:

$$(x_0+h)^2 - (x_0-h)^2 = 4hx_0$$

and for the second, x_0 cancels with x_0 while $h + h$ gives $2h$:

$$(x_0+h) - (x_0-h) = 2h$$

Simplifying

$$= \frac{a \cdot 4hx_0 \; + \; b \cdot 2h}{2h} = \frac{2h\,(2ax_0 + b)}{2h} = 2ax_0 + b$$

taking $2h$ out as a common factor and cancelling it.

Slope of a quadratic function

If

$$f(x) \;=\; ax^2 + bx + c, \qquad a \neq 0,$$

then the slope at the point x is

$$\boxed{g(x) \;=\; 2ax + b}$$

Consistency check

For the prototype $y = x^2$ we have $a = 1$ and $b = 0$, giving $g(x) = 2x$ — exactly what we obtained above.

6 When the Slope Is Zero

The slope is the rate of change. So it is natural to ask: at a point where the slope equals 0, what is the consequence?

Set $g(x) = 0$:

$$2ax + b \;=\; 0 \qquad \Longrightarrow \qquad 2ax \;=\; -b \qquad \Longrightarrow \qquad x \;=\; -\frac{b}{2a}$$

which is legitimate since $a \neq 0$.

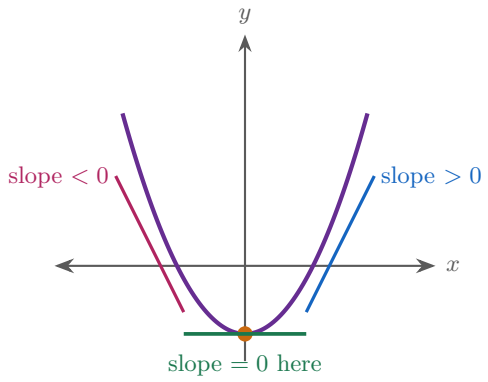
What this number is

This is nothing but the **vertex** of the parabola — the point at which the quadratic function attains its minimum or maximum.

The physical interpretation

The rate of change has come to 0. Prior to that point the rate of change might be positive, and after it negative — so the rate of change is switching over. That is why the point where the slope is 0 gives the minimum or maximum of the original function f.

This is an important application, and it reappears when one studies **calculus**: points of maxima and minima can be derived easily from the expression for the slope.



$x = -\frac{b}{2a}$ is where the rate of change switches sign

Lecture 6 — Quadratic Functions: A Summary

The three-point graphing recipe, the slope $2ax + b$, and the characterisation of maxima and minima

What this lecture covers

We summarise the topics covered on quadratic functions: how the idea generalises the straight line, the cookbook recipe for graphing a parabola from three points, the slope of a quadratic function, and the characterisation of maximum and minimum values.

1 How the Idea Generalises the Line

When we began, we took a straight line of the form

$$f(x) \;=\; mx + c, \qquad m \neq 0$$

and extended the notion to

$$f(x) \;=\; ax^2 + bx + c, \qquad a \neq 0.$$

The analogy

The condition $m \neq 0$ is what makes the term in x meaningful — the function is genuinely **linear** and not a constant. In the same way, the condition $a \neq 0$ ensures that the **quadratic term** x^2 keeps its relevance. Any function of this form is called a **quadratic function**.

2 The Three-Point Recipe for Graphing

Cookbook recipe

1. Find the **axis of symmetry**
$$x \;=\; -\frac{b}{2a}.$$
2. Find the y**-intercept** of the function.
3. Choose any arbitrary point x_1, compute y_1 by substituting into the function, and then use symmetry to obtain a second point x_2 with
$$y_2 \;=\; y_1,$$
since the two lie at equal distances on either side of the axis of symmetry.

Plot these points and join them with a smooth curve.

Graphing a quadratic function is that simple.

3 An Illustration

Graph $f(x) = x^2 + 4x + 4$

Here

$$a = 1, \qquad b = 4, \qquad c = 4.$$

Step 1 — axis of symmetry.

$$x \;=\; -\frac{b}{2a} \;=\; -\frac{4}{2 \times 1} \;=\; -2$$

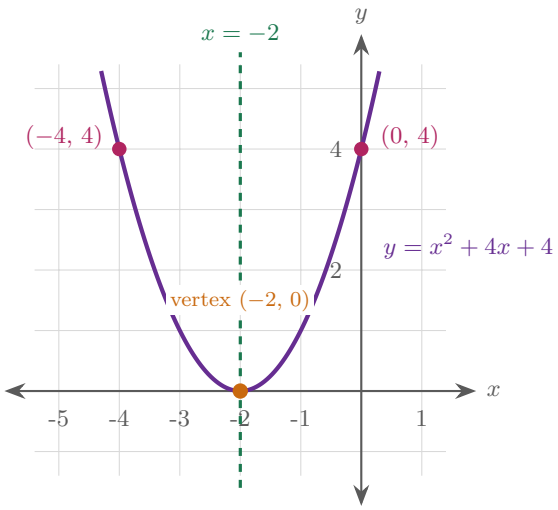
so we draw a vertical line at $x = -2$.
Step 2 — y-intercept. Substituting $x = 0$ gives 4, so we have the point $(0,\ 4)$.
The vertex. Substituting $x = -2$:

$$(-2)^2 + 4(-2) + 4 \;=\; 4 - 8 + 4 \;=\; 0$$

so $(-2,\ 0)$ lies on the curve.
Step 3 — the mirror point. Take $(x_1, y_1) = (0, 4)$. Since $x = -2$ is the axis of symmetry, the matching point lies two units away on the other side, at $x_2 = -4$, with the same value $y_2 = 4$.

So we have three points that suffice to draw the graph:

$$(-4,\ 4), \qquad (-2,\ 0), \qquad (0,\ 4)$$



Joining the three points with a smooth, symmetric curve gives the graph, and the behaviour on both arms follows from the symmetry.

4 The Slope of the Function

We covered the slope of a quadratic function in two ways: once using the angle of inclination — taking tan of the inclination to find the slope at a specific point — and once by generalising the slope of a straight line.

The definition used

$$g(x) \;=\; \frac{f(x+h) - f(x-h)}{2h}$$

the difference between two points either side, divided by the distance between them along the X axis.

A word of caution

This is **not** the general definition of slope — it is the definition specific to quadratic functions, that is, the generalisation of the definition for a straight line. More intricacies are needed to define the slope of a general function; here it does no harm.

For $f(x) = ax^2 + bx + c$ we showed that

$$g(x) \;=\; 2ax + b$$

Applying it to our example

With $a = 1$ and $b = 4$:

$$g(x) \;=\; 2x + 4$$

Setting $g(x) = 0$:

$$2x + 4 \;=\; 0 \qquad \Longrightarrow \qquad x \;=\; -2$$

which is exactly the point where the function attains its minimum — the vertex we already found.

What a zero slope signals

When the slope equals 0 the function is switching over — from an increasing trend to a decreasing one, or from decreasing to increasing. Whatever we wish to learn about the turning behaviour of the function is given at the point where the slope is 0.

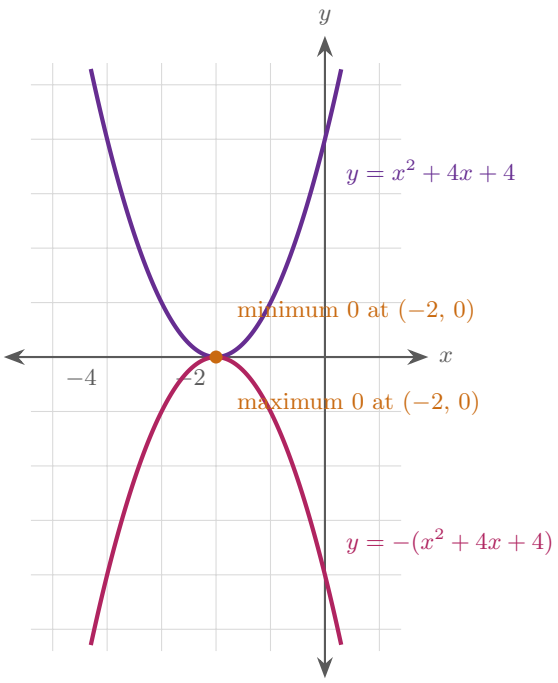
Since $x = -\dfrac{b}{2a}$ is always the vertex of a quadratic function, the function attains either its minimum or its maximum value when evaluated at the vertex. In our example $x = -2$ is the vertex, and the minimum value attained is 0.

5 Reflecting the Curve

What if we simply multiply the function by a negative sign?

From $x^2 + 4x + 4$ to $-(x^2 + 4x + 4)$

Attaching the negative sign **reflects the curve about the X axis**. The function looks exactly the same, as a mirror image, and so the value 0 — previously the minimum — becomes the **maximum** value attained.



Why one has a minimum and the other a maximum

For the upward curve, as $x \to +\infty$ the function goes to $+\infty$, and by symmetry as $x \to -\infty$ it again goes to $+\infty$ — so it never attains a maximum, only a minimum.
For the reflected curve the behaviour is the opposite: it runs off to $-\infty$ in both directions, so it never attains a minimum, only a maximum.

Characterisation of maximum and minimum

$$a > 0 \implies \text{the function always attains a } \textbf{minimum} \text{ value}$$

$$a < 0 \implies \text{the function always attains a } \textbf{maximum} \text{ value}$$

Lecture 7 — Solving by Factoring

Intercept form, the FOIL method, and converting between the standard and intercept forms

What this lecture covers

Previously we found the roots of a quadratic equation by graphing the associated function. Here we meet a well-known algebraic method — **factoring**. For that we introduce a new form of a quadratic function, the **intercept form**, learn the **FOIL** rule for multiplying two binomials, and use it to move between the standard form and the intercept form in both directions.

1 The Intercept Form

Intercept form

If $y = f(x)$, a quadratic function, is written as

$$y \;=\; a\,(x-p)(x-q)$$

then this is called the **intercept form**. The factors $(x-p)$ and $(x-q)$ are called **binomials**, and p and q are the x**-intercepts** of the quadratic function.

In effect, we have located the two x-intercepts on the graph and written the function in terms of them. Whenever such an expression is possible, we may write it this way.

An example

$y = 3(x-1)(x-5)$

Here we are told that 1 and 5 are the x-intercepts — so two values are known, and we can find a third.
At $x = 1$ the value is 0; at $x = 5$ the value is 0.

Since both 1 and 5 give the value 0, there must be an axis of symmetry between them. It lies at the **average** of the two points:

$$x \;=\; \frac{1+5}{2} \;=\; 3$$

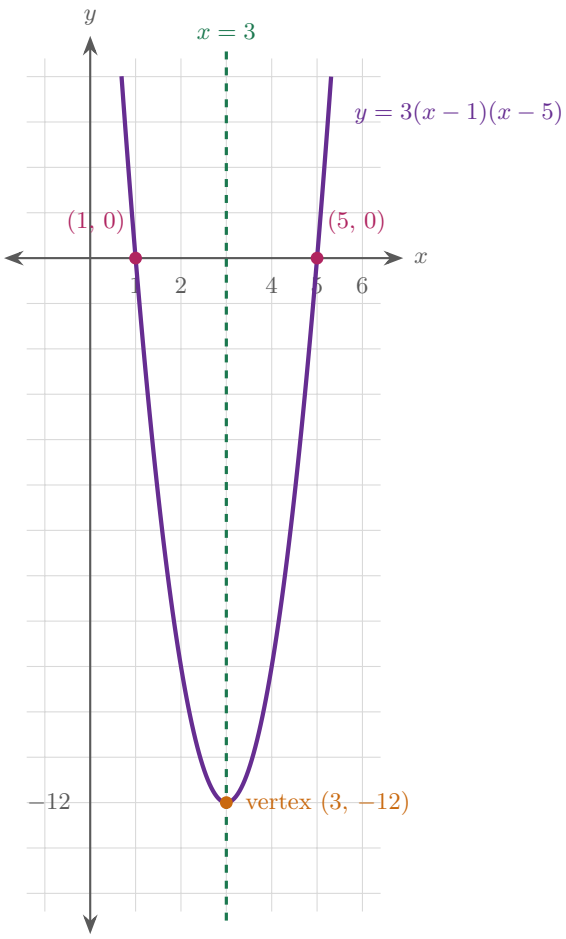
Substituting $x = 3$ into the equation:

$$y \;=\; 3\,(3-1)(3-5) \;=\; 3\times 2\times(-2) \;=\; -12$$

So we have three points:

$$(1,\ 0), \qquad (5,\ 0), \qquad (3,\ -12)$$

and the graph can be plotted easily by joining them with a smooth curve.



2 The FOIL Method

The natural question is: given such an expression, how do we convert it into the **standard form**? Simply by multiplying the two binomials — and for that we use the **FOIL** method.

FOIL

The product of two binomials is the sum of the products of the **F**irst terms, the **O**uter terms, the **I**nner terms and the **L**ast terms.

Taking a general product of two binomials — written with Greek letters, to keep them clear of the a, b, c of the standard form:

$$(\alpha x + \beta)(\gamma x + \delta)$$

First $\alpha x \cdot \gamma x = \alpha\gamma x^2$

Outer $\alpha x \cdot \delta = \alpha\delta x$

Inner $\beta \cdot \gamma x = \beta\gamma x$

Last $\beta \cdot \delta = \beta\delta$

$$(\alpha x + \beta)(\gamma x + \delta) = \alpha\gamma\, x^2 + (\alpha\delta + \beta\gamma)\, x + \beta\delta$$

So $\alpha\gamma$ is the coefficient of x^2, $\alpha\delta + \beta\gamma$ is the coefficient of x, and $\beta\delta$ is the constant term.

A crucial observation

Take the product of the coefficient of x^2 and the constant term:

$$(\alpha\gamma)(\beta\delta) = \alpha\beta\gamma\delta$$

Now take the product of the two pieces making up the coefficient of x:

$$(\alpha\delta)(\beta\gamma) \;=\; \alpha\beta\gamma\delta$$

Both are $\alpha\beta\gamma\delta$. This observation is what we shall use to convert a standard form into an intercept form and back again.

3 Intercept Form → Standard Form

Write a quadratic equation with roots $\frac{2}{3}$ and -4 in standard form

Recall that the standard form is $ax^2+bx+c=0$ with a,b,c integers and $a\neq 0$.
Using the intercept form, since $\frac{2}{3}$ and -4 are the roots:

$$\left(x-\tfrac{2}{3}\right)(x+4) \;=\; 0$$

but this is not yet in standard form.

Applying FOIL with $\alpha=1$, $\beta=-\frac{2}{3}$, $\gamma=1$, $\delta=4$:

$$\text{First: } 1\cdot 1=1 \quad\to\quad x^2$$

$$\text{Outer + Inner: } (1)(4)+\left(-\tfrac{2}{3}\right)(1) \;=\; 4-\tfrac{2}{3} \;=\; \tfrac{10}{3}$$

$$\text{Last: } \left(-\tfrac{2}{3}\right)(4) \;=\; -\tfrac{8}{3}$$

$$x^2+\frac{10}{3}x-\frac{8}{3} \;=\; 0$$

Is this in standard form? No — the coefficients must all be integers. Multiplying **throughout by 3**:

The answer

$$3x^2+10x-8 \;=\; 0$$

4 Standard Form → Intercept Form

Now we run FOIL in reverse.

Convert $5x^2-13x+6$ to intercept form

Write $5x^2-13x+6=(\alpha x+\beta)(\gamma x+\delta)$ and compare with $\alpha\gamma\,x^2+(\alpha\delta+\beta\gamma)x+\beta\delta$:

$$\alpha\gamma \;=\; 5, \qquad \alpha\delta+\beta\gamma \;=\; -13, \qquad \beta\delta \;=\; 6$$

From the observation above,

$$\alpha\beta\gamma\delta \;=\; 5\times 6 \;=\; 30$$

so we must find two numbers, $\alpha\delta$ and $\beta\gamma$, whose **product is** 30 and whose **sum is** -13.

Using prime factorisation

Recall the prime factorisation from Week 1:

$$30 \;=\; 2\times 3\times 5$$

Combining 5 and 2 gives 10, and $10 + 3 = 13$ — but we need -13. Since the product of two negative numbers is positive, take

$$(-10) \times (-3) \;=\; 30, \qquad (-10) + (-3) \;=\; -13 \qquad \checkmark$$

so $\alpha\delta = -10$ and $\beta\gamma = -3$ (or the other way round — it does not matter).

Splitting and grouping

Rewrite the middle term using these two numbers:

$$5x^2 - 10x - 3x + 6$$

Take the greatest common factor out of the first two terms, and again out of the last two:

$$5x\,(x-2) \;-\; 3\,(x-2)$$

The brackets $(x-2)$ are the same, so

$$(5x-3)(x-2)$$

Is this the intercept form? Not yet — the intercept form is $a(x-p)(x-q)$. Dividing the first bracket by 5 and taking the 5 outside:

The intercept form

$$5\left(x - \frac{3}{5}\right)(x-2)$$

5 Three More Examples

Example 1 — $x^2 = 8x$

Here no FOIL method is really needed. Rearranging and taking out the greatest common factor:

$$x^2 - 8x \;=\; 0 \qquad \Longrightarrow \qquad x\,(x-8) \;=\; 0$$

Reading this as $(x-0)(x-8) = 0$, the roots are

$$x = 0 \qquad \text{and} \qquad x = 8.$$

This settles the simple case where the constant term is absent.

Example 2 — $x^2 - 4x + 4 = 0$

Here $\alpha\beta\gamma\delta = 4$ and $\alpha\delta + \beta\gamma = -4$. The factor pairs of 4 are 1×4 and 2×2; only the second can give the sum -4, and for that both must be negative:

$$(-2) \times (-2) = 4, \qquad (-2) + (-2) = -4$$

Substituting into the master equation:

$$x^2 - 2x - 2x + 4 \;=\; 0$$

Taking out the greatest common factors:

$$x\,(x-2) - 2\,(x-2) \;=\; 0 \qquad \Longrightarrow \qquad (x-2)^2 \;=\; 0$$

So 2 is the **repeated** root of this equation.

Example 3 — $x^2 - 25 = 0$

This one is interesting: 25 is a perfect square, 5^2. Here

$$\alpha\beta\gamma\delta \;=\; -25, \qquad \alpha\delta + \beta\gamma \;=\; 0$$

The factor pairs of 25 are 1×25 and 5×5; only the second can give the sum 0. Since the product is -25, one factor carries a plus sign and the other a minus sign: $\alpha\delta = +5$ and $\beta\gamma = -5$ (or the reverse). Substituting:

$$x^2 - 5x + 5x - 25 \;=\; 0$$

Taking out the greatest common factors x and 5 respectively:

$$x\,(x-5) + 5\,(x-5) \;=\; 0 \qquad \Longrightarrow \qquad (x+5)(x-5) \;=\; 0$$

so the roots are $x = 5$ and $x = -5$.

Why the intercept form is the goal

In every one of these examples the final expression is written in **intercept form**. Once an expression is in that form, finding the roots is very easy — in fact, once it is written in intercept form, the roots have *already* been found.

On practice

This concept is quite intricate, so it needs practice — solve as many problems as you can.

Lecture 8 — Completing the Square

A second algebraic method for the roots, its caution, and the link to the quadratic formula

What this lecture covers

We learn another method for finding the roots of a quadratic equation — **completing the square** — which also has a very good connection with the well-known **quadratic formula**, explored later. The method is demonstrated through an example, starting from the factoring approach already seen.

1 The Old Method, Recalled

Solving $x^2 + 10x - 24 = 0$ by factoring

Here the product of the leading coefficient and the constant term is

$$\alpha\beta\gamma\delta \;=\; 1 \times (-24) \;=\; -24, \qquad \alpha\delta + \beta\gamma \;=\; 10.$$

Using prime factorisation and rearranging the factors of -24 so that the sum is 10, one such choice is

$$\alpha\delta = 12, \qquad \beta\gamma = -2.$$

Substituting for the coefficient of x:

$$x^2 + 12x - 2x - 24 \;=\; 0$$

Taking out the greatest common factors:

$$x\,(x+12) - 2\,(x+12) \;=\; 0 \qquad \Longrightarrow \qquad (x+12)(x-2) \;=\; 0$$

so the roots are

$$x = -12 \qquad \text{and} \qquad x = 2.$$

2 The New Idea

Somebody then asked: why should I bother about that last term? It is just a constant — so let me move it across and work on what is left. From that thought comes the method of **completing the square**.

Rewrite the equation as

$$x^2 + 10x \;=\; 24$$

The next question is: looking at $x^2 + 10x$, is there something we can add to make it a **complete square**?

What a complete square means

$$(x+a)^2 \;=\; x^2 + 2ax + a^2$$

So to turn $x^2 + 10x$ into such a square we must add a suitable number — and to keep the

equation balanced, the same number must be added on **both sides**.

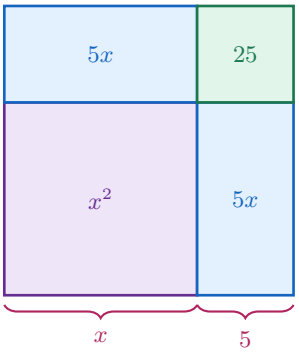
3 Finding the Number to Add

Which number is it? Compare the middle terms:

$$10 = 2a \qquad \Longrightarrow \qquad a = 5$$

so the number to be added is

$$a^2 = 5^2 = 25.$$



$x^2 + 10x + 25 = (x+5)^2$

the added 25 completes the square

4 Carrying It Through

Add 25 to both sides:

$$x^2 + 10x + 25 = 24 + 25 = 49$$

The left-hand side is now a complete square, and in this case the right-hand side happens to be a perfect square too — though that need not always be so:

$$(x+5)^2 = 7^2$$

Why only two cases, not four

Both sides are squares, so strictly one could write

$$+(x+5) = \pm 7 \qquad \text{and} \qquad -(x+5) = \pm 7$$

which looks like four cases. But the second pair reduces to the same two equations as the first, so it suffices to consider

$$x + 5 = \pm 7.$$

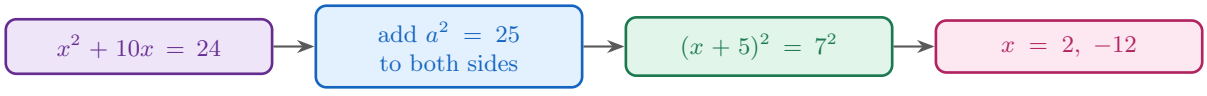
Subtracting 5 from both sides:

$$x = -5 + 7 = 2 \qquad x = -5 - 7 = -12$$

The roots

$$x = 2 \qquad \text{and} \qquad x = -12$$

These match exactly the roots obtained by factoring — the solution set is the same.



5 Why One Might Prefer This Method

Which method to use is now a personal choice — but note what the new one avoids.

No factorisation needed

Suppose the constant were not 24 but some awkward number, so that factoring — finding the prime factorisation and hunting for two numbers with the right product and sum — became difficult.
In completing the square we never use that property at all. So we can dispense with it entirely and need not worry about it.

A note of caution

This method cannot be used when, after adding a^2, the number on the right-hand side is **negative** — because the square root of a negative number is not defined.
In that case, according to this method, there is no real solution.

right-hand side ≥ 0
take $\pm\sqrt{\ }$ and solve

right-hand side < 0
no real solution by this method

So the method has some limitations, but it is quite powerful for solving problems. We shall see how to overcome those limitations, and the beautiful connection with the **quadratic formula**, in the next few lectures.

Lecture 9 — The Quadratic Formula and the Discriminant

Irrational roots, deriving the formula, the discriminant table, and why the axis of symmetry is $x = -\frac{b}{2a}$

What this lecture covers

We continue with completing the square when the right-hand side is *not* a perfect square, then use the same method on a general quadratic to derive the **quadratic formula**. The expression under the square root turns out to be the **discriminant**, which tells us the number and type of roots. We close by comparing all four methods and finally deriving the axis of symmetry.

1 When the Right-Hand Side Is Not a Perfect Square

Solve $x^2 - 4x + 4 = 32$

The first step is already done — the left-hand side is a complete square:

$$(x-2)^2 = 32$$

and 32 is *not* a perfect square.

Proceeding in the same manner, take the square root of both sides:

$$x - 2 = \pm\sqrt{32}$$

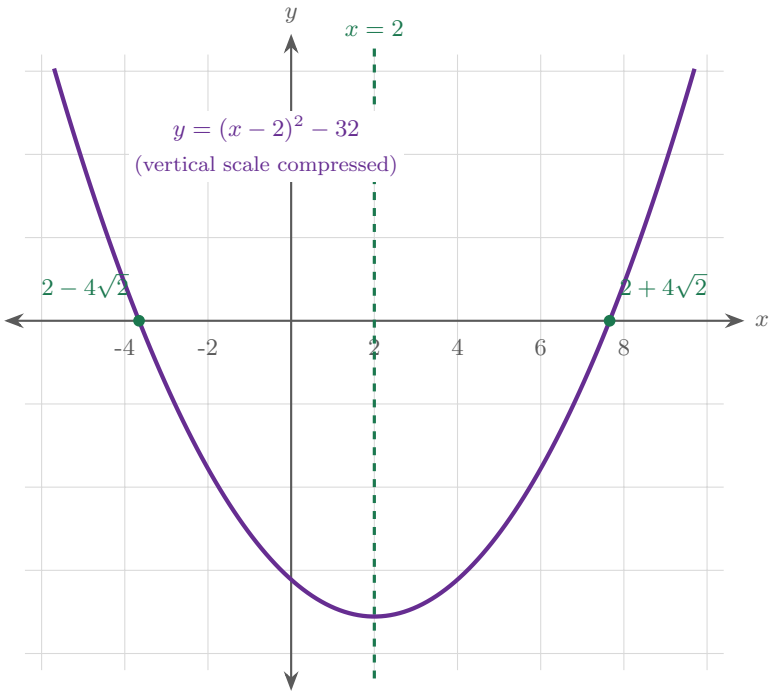
and decompose 32:

$$\sqrt{32} = \sqrt{16 \times 2} = 4\sqrt{2}$$

The roots

$$x = 2 \pm 4\sqrt{2}$$

These are **irrational roots**, since $\sqrt{2}$ is an irrational number.



Verifying with the graph makes clear exactly where $2 \pm 4\sqrt{2}$ sit: the two green dots mark the locations of the roots.

2 Deriving the Quadratic Formula

Now apply completing the square to a **general** quadratic equation.

$$ax^2 + bx + c \;=\; 0, \qquad a \neq 0$$

Since $a \neq 0$ we may divide throughout by a:

$$x^2 + \frac{b}{a}\,x + \frac{c}{a} \;=\; 0$$

Following the method, push the constant $\dfrac{c}{a}$ to the other side, where it takes a negative sign:

$$x^2 + \frac{b}{a}\,x \;=\; -\frac{c}{a}$$

The number to add

Comparing with $x^2 + 2kx + k^2 = (x + k)^2$, the middle coefficient $\dfrac{b}{a}$ must equal $2k$, so $k = \dfrac{b}{2a}$. The number to add on both sides is therefore k^2:

$$\left(\frac{b}{2a}\right)^2 \;=\; \frac{b^2}{4a^2}$$

$$x^2 + \frac{b}{a}\,x + \frac{b^2}{4a^2} \;=\; \frac{b^2}{4a^2} - \frac{c}{a}$$

Recognising the square

The left-hand side is now $\left(x + \frac{b}{2a}\right)^2$. On the right, taking $4a^2$ as the common denominator — multiplying $\frac{c}{a}$ above and below by $4a$:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

Taking the square root of both sides:

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

The quadratic formula

$$\boxed{x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}}$$

So the same method of completing the square gives a general formula, known as the **quadratic formula**.

3 The Discriminant

Discriminant

The term under the square root,

$$D = b^2 - 4ac,$$

is called the **discriminant** — because it *discriminates* between the possible kinds of roots.

$$b^2 - 4ac > 0 \implies \text{two real roots}$$
$$b^2 - 4ac = 0 \implies \text{one repeated root}$$
$$b^2 - 4ac < 0 \implies \text{no real roots}$$

In the last case we would be taking the square root of a negative number, which goes into the complex domain.

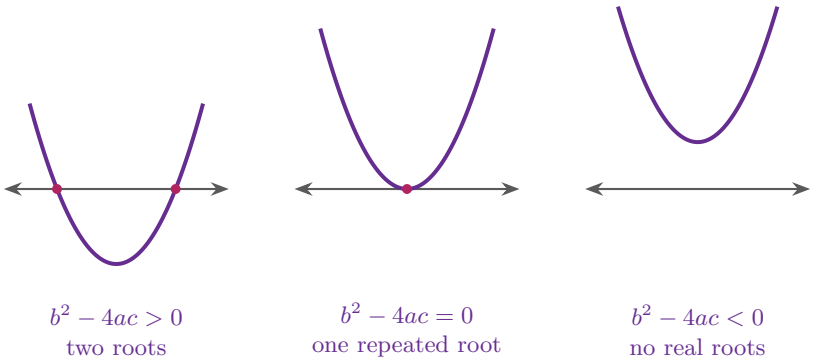
The full table

Value of the discriminant	Number and type of roots
$b^2 - 4ac > 0$ and a perfect square	two real **rational** roots
$b^2 - 4ac > 0$ but not a perfect square	two real **irrational** roots
$b^2 - 4ac = 0$	one real **rational** root (repeated)
$b^2 - 4ac < 0$	**no** real roots

Recall from Week 1 that the real line splits into rationals and irrationals — that is the splitting being used here.

A condition for the rational / irrational distinction

These rows assume that a, b, c are **rational numbers**. If they are not, we simply suppress that distinction: there will still be two real roots when $D > 0$, but we cannot say whether they are rational or irrational.
For instance, $(x - \pi)^2 = 0$ has $D = 0$ yet its root is π, an irrational number. If we do not wish to distinguish rational from irrational, the condition is not needed — a, b, c may be any real numbers, with the only standing condition being $a \neq 0$.



$b^2 - 4ac > 0$ two roots | $b^2 - 4ac = 0$ one repeated root | $b^2 - 4ac < 0$ no real roots

The third case is exactly our earlier example $x^2 + 1$, which never touches the X axis.

4 Two Worked Examples

Find the discriminant and describe the roots

Equation 1: $9x^2 - 12x + 4 = 0$, so $a = 9$, $b = -12$, $c = 4$:

$$b^2 - 4ac \;=\; (-12)^2 - 4(9)(4) \;=\; 144 - 144 \;=\; 0$$

By the table, it has **one repeated rational root**.
Equation 2: $2x^2 + 16x + 33 = 0$, so $a = 2$, $b = 16$, $c = 33$:

$$b^2 - 4ac \;=\; 16^2 - 4(2)(33) \;=\; 256 - 264 \;=\; -8$$

Since $b^2 - 4ac < 0$, it has **no real roots**.

5 Comparing the Four Methods

Method	When to use it
Graphing	Unless the solutions are integers it will not give a good result, but it is the best method to *verify* results found algebraically — a calculation mistake is revealed very easily. It can also be used occasionally to find roots.
Factoring	Not helpful when the factors are not easily visible — imagine a constant term of 26.2. But it is very helpful when the constant term is 0, or when the numbers are nice, such as 49 or 24, and the factors can be guessed.
Completing the square	Always works when real roots exist, and is very easy when b is even. A negative right-hand side is not a failure of the method — it is the method telling us there are *no* real roots, the complex case being outside this course.
Quadratic formula	Derived from completing the square, and gives the answer all the time.

The dependable pair

For our purposes, **completing the square** and the **quadratic formula** will always give the answer, irrespective of whether the coefficients are rational, irrational or otherwise awkward numbers.

6 Why the Axis of Symmetry Is $x = -\frac{b}{2a}$

We have claimed all along that $x = -\dfrac{b}{2a}$ is the axis of symmetry, without proof. The derivation is an application of completing the square.

Take a general quadratic function

$$f(x) \;=\; ax^2 + bx + c, \qquad a \neq 0$$

and pull a out as a common factor:

$$f(x) \;=\; a\left[x^2 + \frac{b}{a}\,x + \frac{c}{a}\right]$$

One difference from before

When solving an equation we pushed $\dfrac{c}{a}$ to the other side. Here there is no other side, so we **retain** it — and instead *add and subtract* the required number $\dfrac{b^2}{4a^2}$ inside the bracket.

$$f(x) \;=\; a\left[x^2 + \frac{b}{a}\,x + \frac{b^2}{4a^2} \;-\; \frac{b^2}{4a^2} + \frac{c}{a}\right]$$

Recognising the first three terms as a complete square:

The vertex form

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

Reading off the symmetry

Look at this equation carefully. It is the same quadratic function written in a different form, and the trailing part

$$c - \frac{b^2}{4a}$$

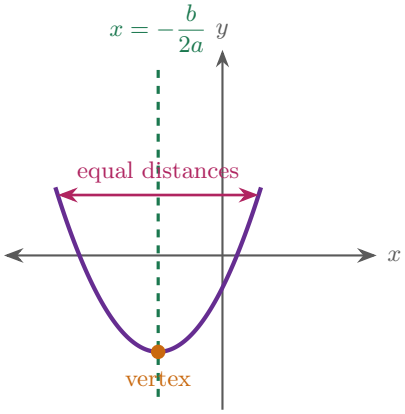
is just **a constant** — it only shifts the curve up or down the y axis. All the symmetry is decided by the squared term.

The conclusion

Just as $y = x^2$ is symmetric about the Y axis, the expression $\left(x + \frac{b}{2a}\right)^2$ is symmetric about the vertical line where the bracket vanishes:

$$x + \frac{b}{2a} = 0 \quad \Longrightarrow \quad x = -\frac{b}{2a}$$

That vertical line is therefore the **axis of symmetry**, which answers the question we had left open.



This ends our topic on quadratic functions and quadratic equations.

Lecture 10 — Representations of a Quadratic Function

Standard form, intercept form and vertex form — and moving between them

What this lecture covers

A quadratic function can be written in more than one way, and each way makes a different feature obvious. We look at the three standard representations — **standard form**, **intercept form** and **vertex form** — see how to move between them, note when a representation is *not* available, and recover the quadratic formula and the coordinates of the vertex along the way.

1 The Three Representations

Standard form
$f(x) = ax^2 + bx + c$
shows the y-intercept c

Intercept form
$f(x) = a(x-p)(x-q)$
shows the roots p and q

Vertex form
$f(x) = a(x-h)^2 + k$
shows the vertex $(h,\ k)$

The standing condition

In every one of these, $a \neq 0$ — otherwise the function is not quadratic at all.

2 Standard Form and Intercept Form

Recall the quadratic equation

$$ax^2 + bx + c \ = \ 0, \qquad a \neq 0,$$

where we ask for the values of x that make the function zero — the **roots**. If we know those roots p and q, the function can be written in **intercept form**:

$$f(x) \ = \ a\,(x-p)(x-q)$$

Moving from the intercept form back to the standard form is done by multiplying out the two binomials with the **FOIL** method, as seen earlier.

An example where it works

$$x^2 + 4x + 4 \ = \ (x+2)(x+2) \ = \ (x+2)^2$$

so the intercept form exists, with $p = q = -2$ — a repeated root.

When the intercept form is not available

Can $f(x) = x^2 + 1$ be written in intercept form?

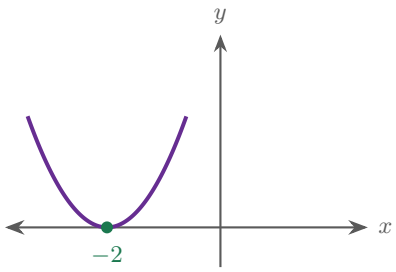
We would need real numbers p and q with

$$x^2 + 1 \ = \ a\,(x-p)(x-q)$$

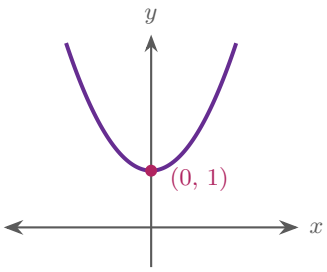
But $x^2 + 1$ has **no real roots** — its smallest value is 1, so the curve never touches the X axis. So no such real p and q exist.

The conclusion

The intercept form is **not always possible**. It exists only when the quadratic actually has real roots — that is, when the discriminant is not negative.



$x^2 + 4x + 4$
intercept form exists



$x^2 + 1$
no real roots, so no intercept form

3 Standard Form to Vertex Form

To obtain the third representation we use **completing the square** — the same technique as before, but applied to the function rather than to an equation.

Start from

$$f(x) = ax^2 + bx + c, \qquad a \neq 0$$

and take a out as a common factor:

$$f(x) = a\left[x^2 + \frac{b}{a}x + \frac{c}{a}\right]$$

The difference from solving an equation

When solving an equation we pushed the constant to the other side. Here there is no other side, so we keep it and instead **add and subtract** the required number inside the bracket.

The number required is $\left(\frac{b}{2a}\right)^2 = \frac{b^2}{4a^2}$:

$$f(x) = a\left[x^2 + \frac{b}{a}x + \frac{b^2}{4a^2} - \frac{b^2}{4a^2} + \frac{c}{a}\right]$$

Recognising the first three terms as a complete square and multiplying the remaining terms by a:

Vertex form

$$f(x) = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$$

so that, comparing with $a(x-h)^2 + k$,

$$h = -\frac{b}{2a}, \qquad k = c - \frac{b^2}{4a} = \frac{4ac - b^2}{4a}$$

The coordinates of the vertex

$$\text{vertex} = \left(-\frac{b}{2a}, \frac{4ac - b^2}{4a} \right)$$

The squared term decides the symmetry; the trailing constant only shifts the curve up or down.

4 Recovering the Quadratic Formula

Setting the vertex form equal to zero recovers the roots. From

$$a\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a}$$

dividing by a:

$$\left(x + \frac{b}{2a}\right)^2 = \frac{b^2 - 4ac}{4a^2}$$

and taking the square root of both sides:

$$x + \frac{b}{2a} = \pm\frac{\sqrt{b^2 - 4ac}}{2a}$$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

The two special cases

Something important happens with the square root of $b^2 - 4ac$:

- If $b^2 - 4ac < 0$ the number under the root is not real, so the equation has **no real solution**.
- If $b^2 - 4ac = 0$ the two values coincide, giving a **repeated** real solution.

5 The Vertex Sits Midway Between the Roots

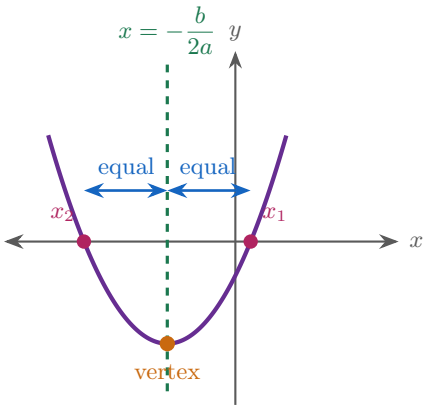
When the two roots exist, call them

$$x_1 = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad x_2 = \frac{-b - \sqrt{b^2 - 4ac}}{2a}.$$

Their **midpoint** is

$$\frac{x_1 + x_2}{2} = \frac{1}{2} \cdot \frac{-2b}{2a} = -\frac{b}{2a}$$

which is exactly the x coordinate of the vertex — the square-root terms cancel. So the vertex lies at the **midpoint of the two roots**, which is another way of seeing why $x = -\dfrac{b}{2a}$ is the axis of symmetry.



6 Putting It Together on One Example

Write $f(x) = x^2 + 2x + 4$ in all three forms

Standard form. $a = 1$, $b = 2$, $c = 4$, so the y-intercept is 4.
Vertex form. Completing the square:

$$x^2 + 2x + 4 \;=\; (x^2 + 2x + 1) - 1 + 4 \;=\; (x + 1)^2 + 3$$

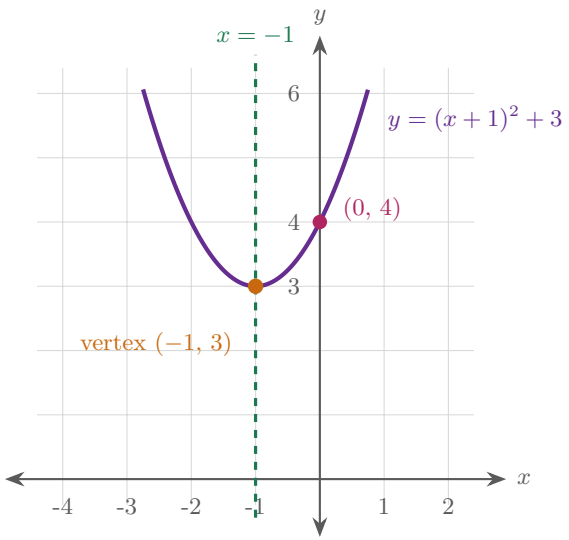
so the vertex is $(-1,\ 3)$, agreeing with

$$h = -\frac{b}{2a} = -1, \qquad k = c - \frac{b^2}{4a} = 4 - 1 = 3.$$

Intercept form. The discriminant is

$$b^2 - 4ac \;=\; 4 - 16 \;=\; -12 \;<\; 0$$

so there are no real roots, and the intercept form does **not** exist for this function.



The curve sits entirely above the X axis, which is exactly what a negative discriminant tells us.

References and Acknowledgement

These notes are based on the video lectures of Mathematics–1 offered by the IIT Madras BS Degree Programme.

Notes created by: Vishal (Instructor)